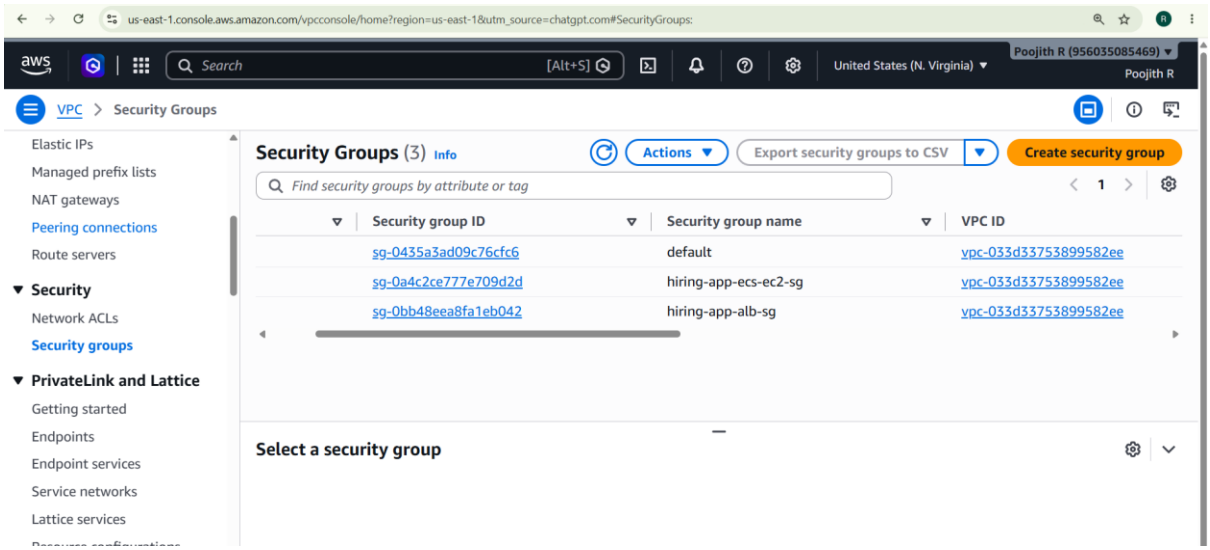
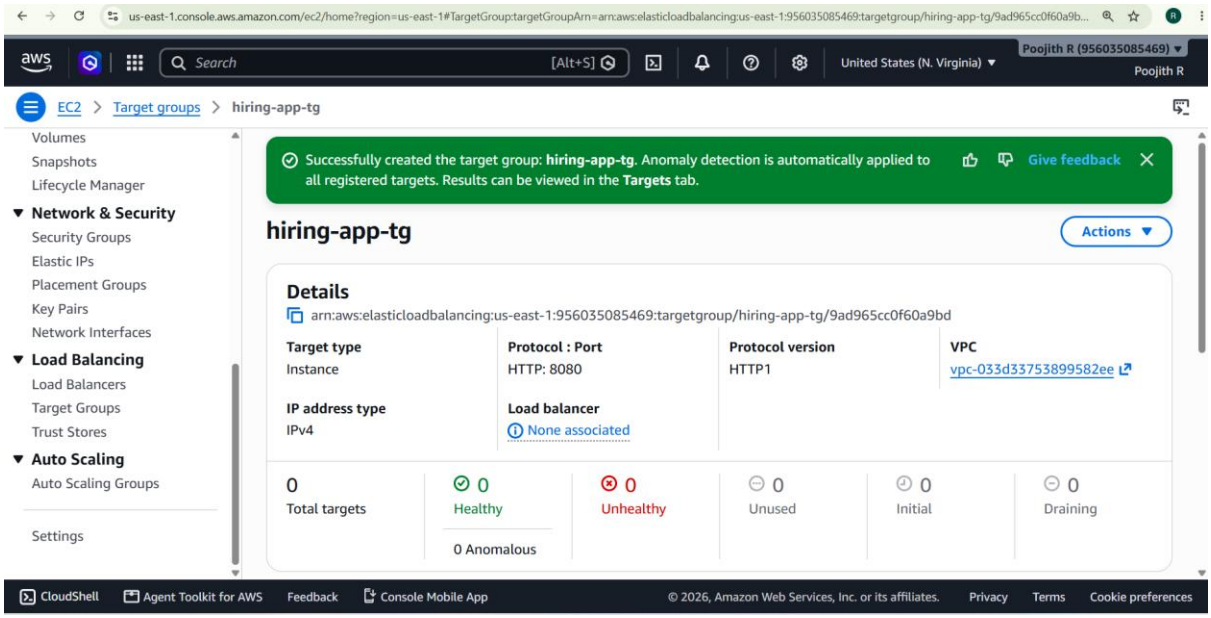
1. Setup a highly available ecs cluster with load balancer and dynamic port mapping.
   NOTE:
   Your cluster should maintain atleast 6 tasks at any point of time and should be highly availble across multiple AZ's.
   Use the below image from deployment:
   sabair0509/hiring-app:works

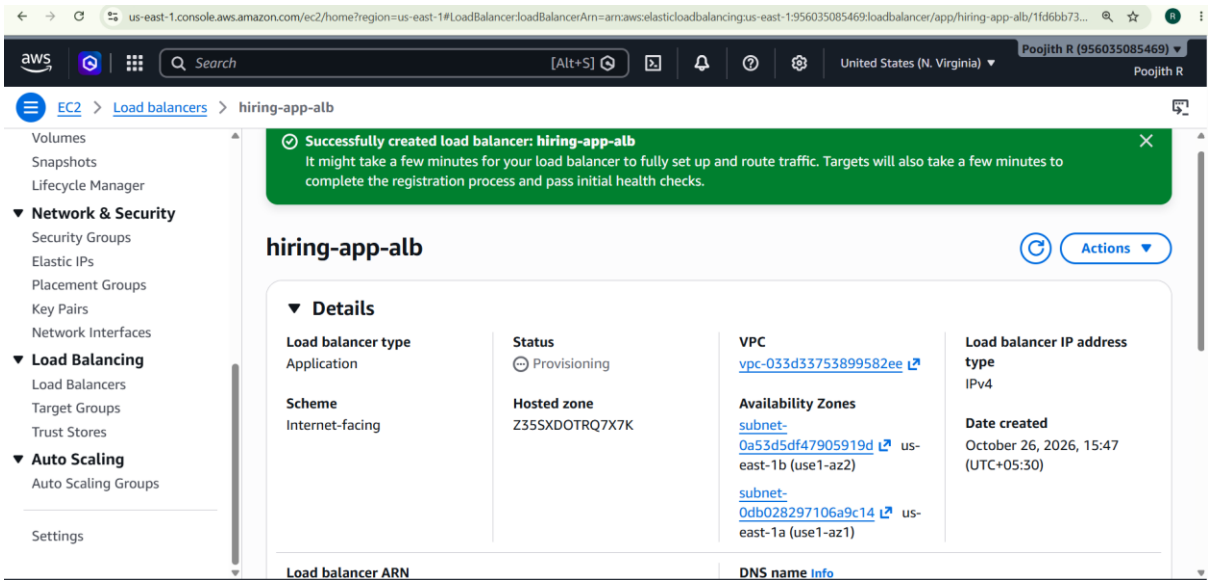
Implementation Steps

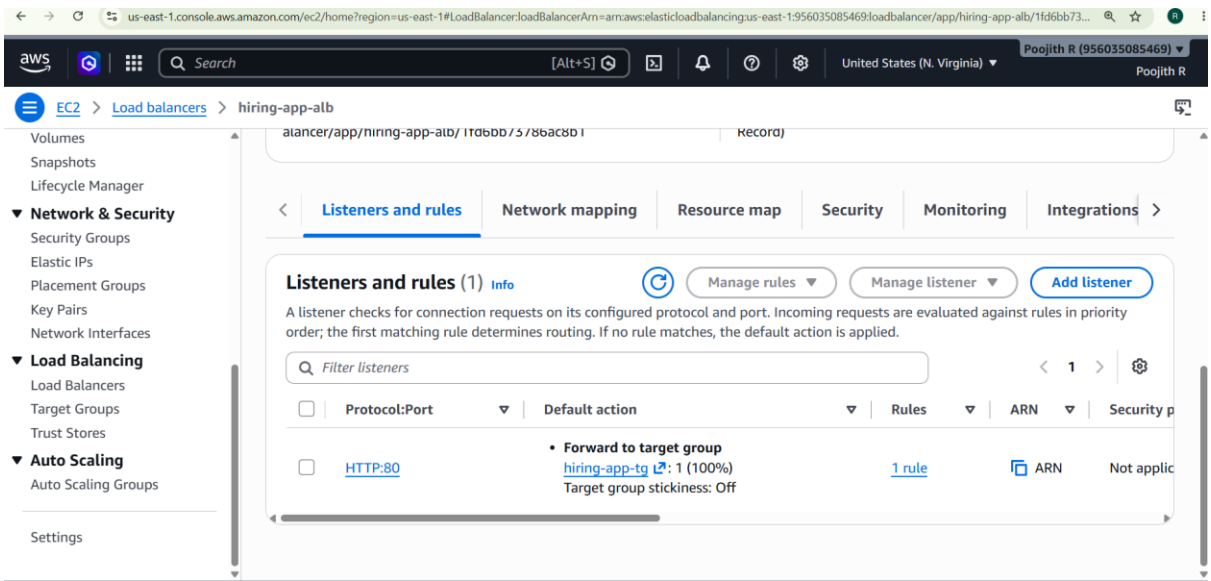
1. Logged in to the AWS Management Console and selected the US East (N. Virginia) us-east-1 region.
2. Selected the default VPC and identified subnets in us-east-1a and us-east-1b.
3. Created an ECS cluster named hiring-app-ha-cluster.
4. Selected Fargate and Self-managed instances and configured a new Auto Scaling group using the On-Demand provisioning model.
5. Selected the Amazon ECS-Optimized Amazon Linux 2023 AMI and configured EC2 instances across both Availability Zones.
6. Configured the ECS instance IAM role and cluster registration settings.
7. Created a task definition named hiring-app-task using the Docker image sabair0509/hiring-app:works.
8. Configured the task definition to use EC2 launch capacity and bridge network mode.
9. Configured container port 8080 and host port 0 to enable dynamic port mapping.
10. Created a security group named hiring-app-alb-sg to allow incoming HTTP traffic on port 80.
11. Created a security group named hiring-app-ecs-ec2-sg to allow application traffic on dynamic host ports 32768–61000 from the ALB security group.
12. Created an internet-facing Application Load Balancer named hiring-app-alb across two Availability Zones.
13. Created a target group named hiring-app-tg with target type Instances and configured the HTTP health check path as /hiring/.
14. Configured the ALB listener on HTTP port 80 to forward incoming requests to hiring-app-tg.

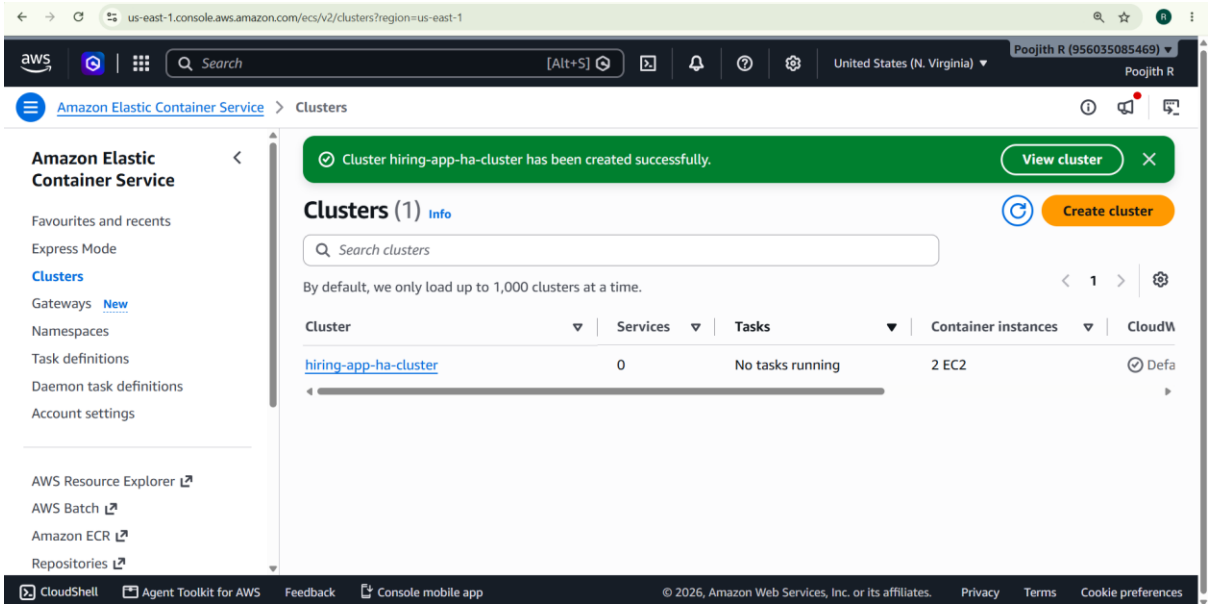
15. Created an ECS service named hiring-app-ha-service using task definition hiring-app-task:3.
16. Initially deployed one task to verify application startup and load balancer health, then updated the desired task count to 6.
17. Configured the task placement strategy to spread tasks across Availability Zones and EC2 instances using AZ balanced spread.
18. Attached the existing ALB listener and target group to the ECS service using container port 8080.
19. Configured ECS service auto scaling with a minimum of 6 tasks and a maximum of 12 tasks, using target tracking based on average CPU utilization with a target value of 60%.
20. Verified that all six ECS tasks were running, with three tasks distributed in us-east-1a and three tasks in us-east-1b.
21. Verified that the target group reported healthy targets and that the application was accessible through the ALB using the /hiring/ application path.
22. Accessed the hiring application through the ALB DNS name and confirmed that the application loaded successfully.

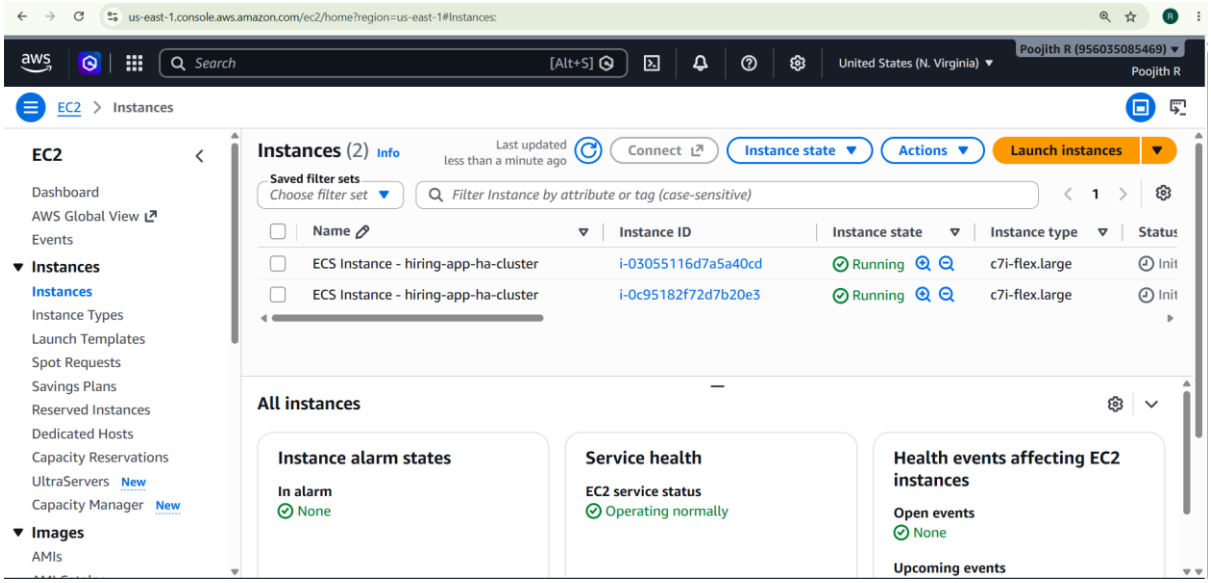


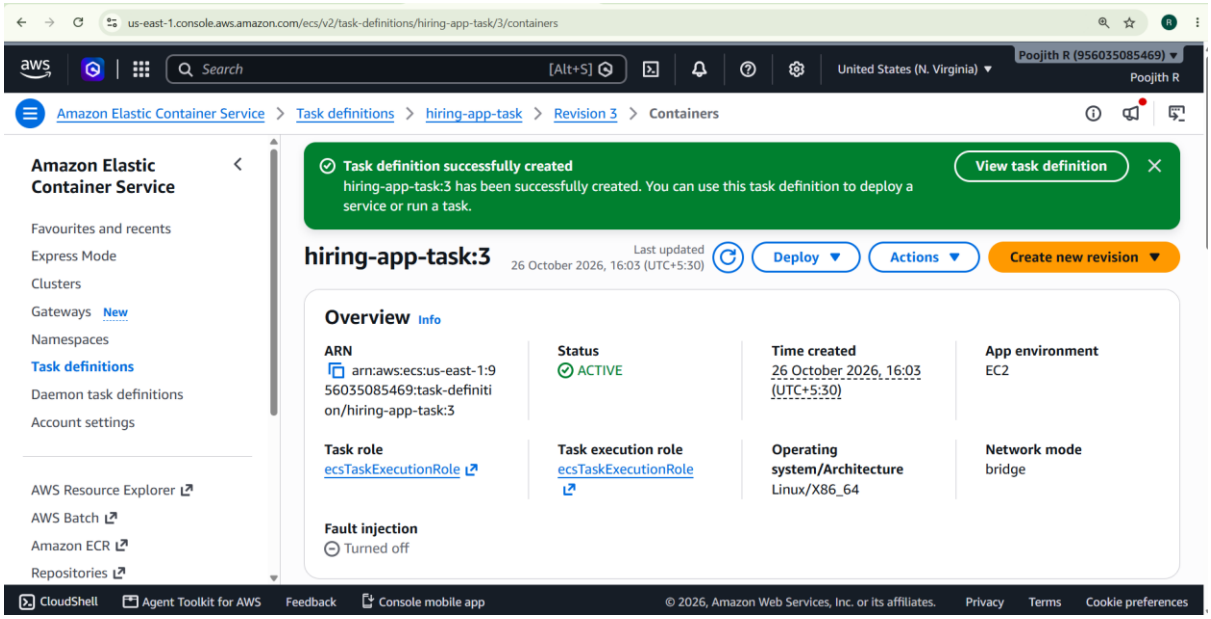

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/hiring-app-tg/9ad965cc0f60a9b...
Search [Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
EC2 > Target groups > hiring-app-tg
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Successfully created the target group: hiring-app-tg. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.
Give feedback
hiring-app-tg
Actions
Details
arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/hiring-app-tg/9ad965cc0f60a9bd
Target type
Instance
Protocol : Port
HTTP: 8080
Protocol version
HTTP1
VPC
vpc-033d33753899582ee
IP address type
IPv4
Load balancer
None associated
0
Total targets
0
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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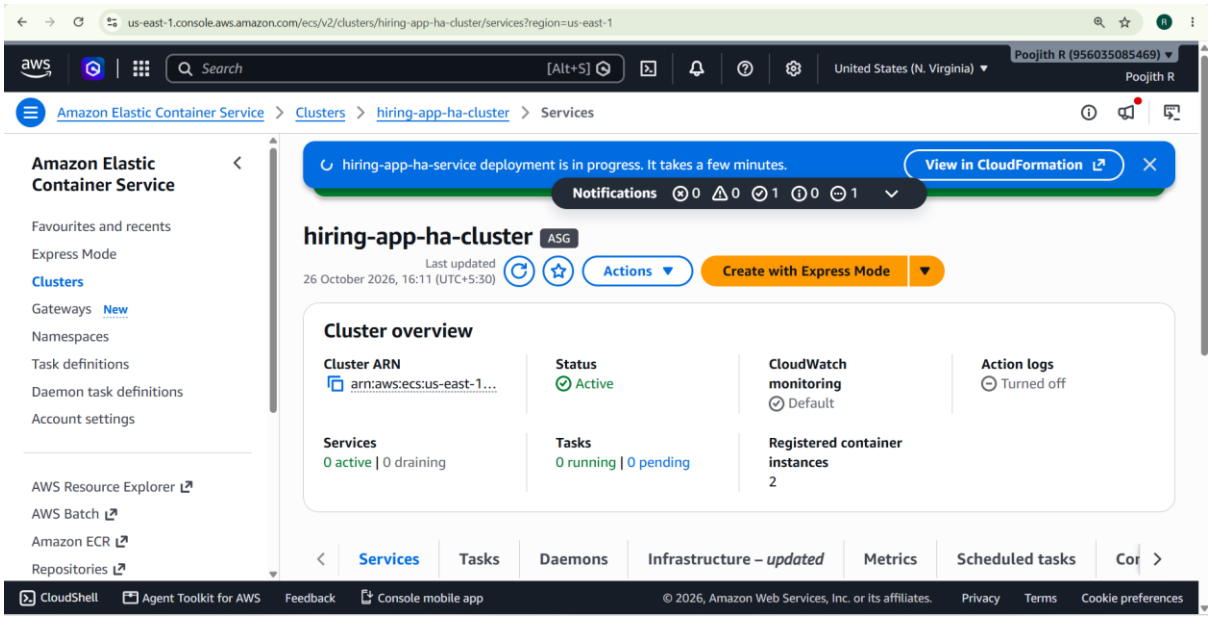

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:us-east-1:956035085469:loadbalancer/app/hiring-app-alb/1fd6bb73...
Search [Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
EC2 > Load balancers > hiring-app-alb
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Successfully created load balancer: hiring-app-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.
hiring-app-alb
Actions
Details
Load balancer type
Application
Status
Provisioning
VPC
vpc-033d33753899582ee
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-0a53d5df47905919d us-east-1b (use1-az2)
subnet-0db028297106a9c14 us-east-1a (use1-az1)
Date created
October 26, 2026, 15:47 (UTC+05:30)
Load balancer ARN
DNS name Info

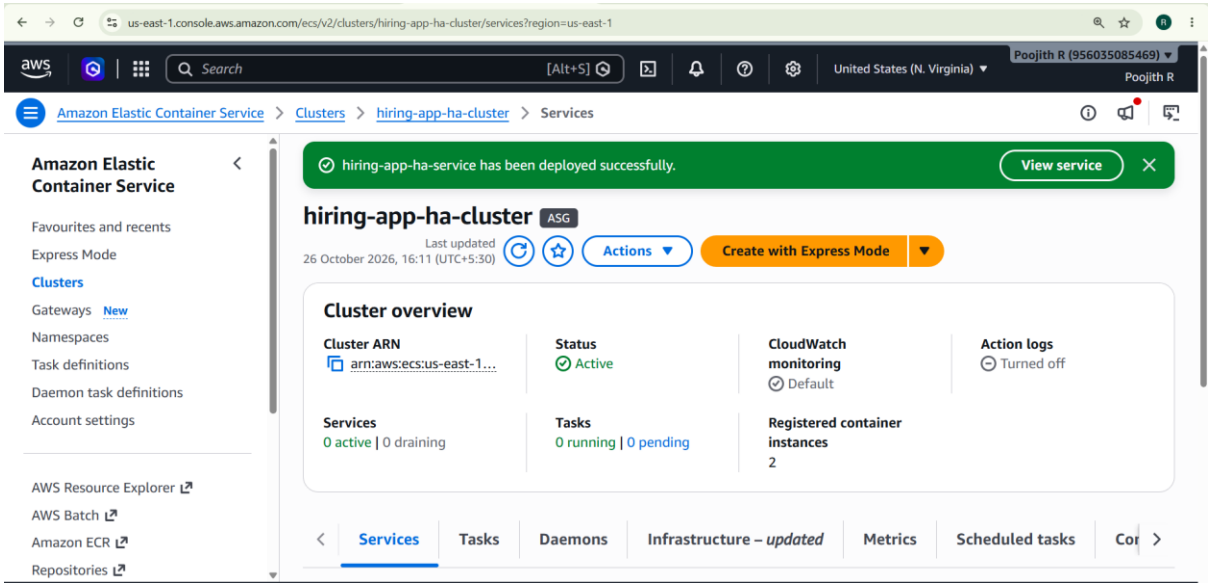

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:us-east-1:956035085469:loadbalancer/app/hiring-app-alb/1fd6bb73...
Search [Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
EC2 > Load balancers > hiring-app-alb
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.
Filter listeners
Protocol:Port
Default action
Rules
ARN
Security p
HTTP:80
Forward to target group
hiring-app-tg: 1 (100%)
Target group stickiness: Off
1 rule
ARN
Not applic

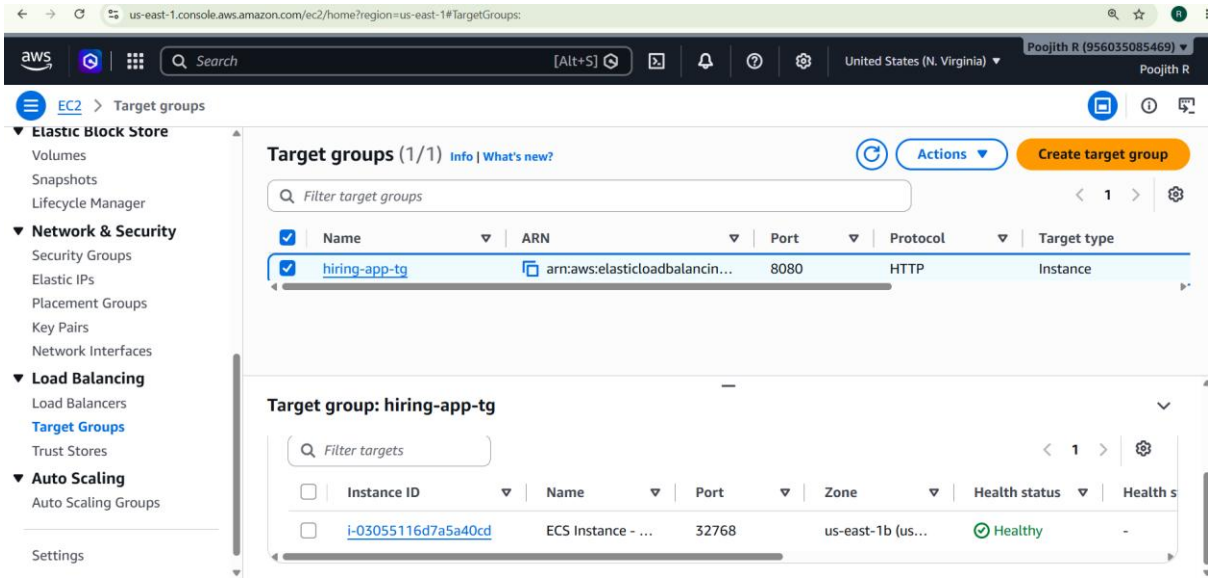

us-east-1.console.aws.amazon.com/ecs/v2/clusters?region=us-east-1
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service > Clusters
Amazon Elastic Container Service
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
Cluster hiring-app-ha-cluster has been created successfully.
View cluster
Clusters (1) Info
Create cluster
Search clusters
By default, we only load up to 1,000 clusters at a time.
Cluster
Services
Tasks
Container instances
hiring-app-ha-cluster
0
No tasks running
2 EC2
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
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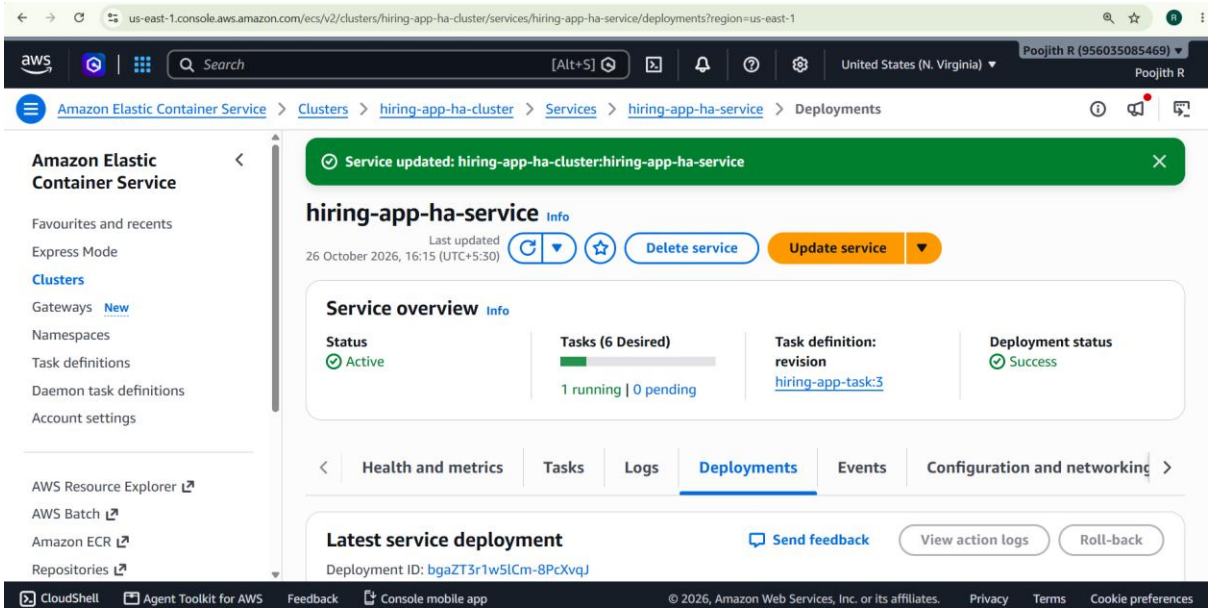

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Instances (2) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
ECS Instance - hiring-app-ha-cluster
i-03055116d7a5a40cd
Running
c7i-flex.large
ECS Instance - hiring-app-ha-cluster
i-0c95182f72d7b20e3
Running
c7i-flex.large
All instances
Instance alarm states
In alarm
None
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events
None
Upcoming events

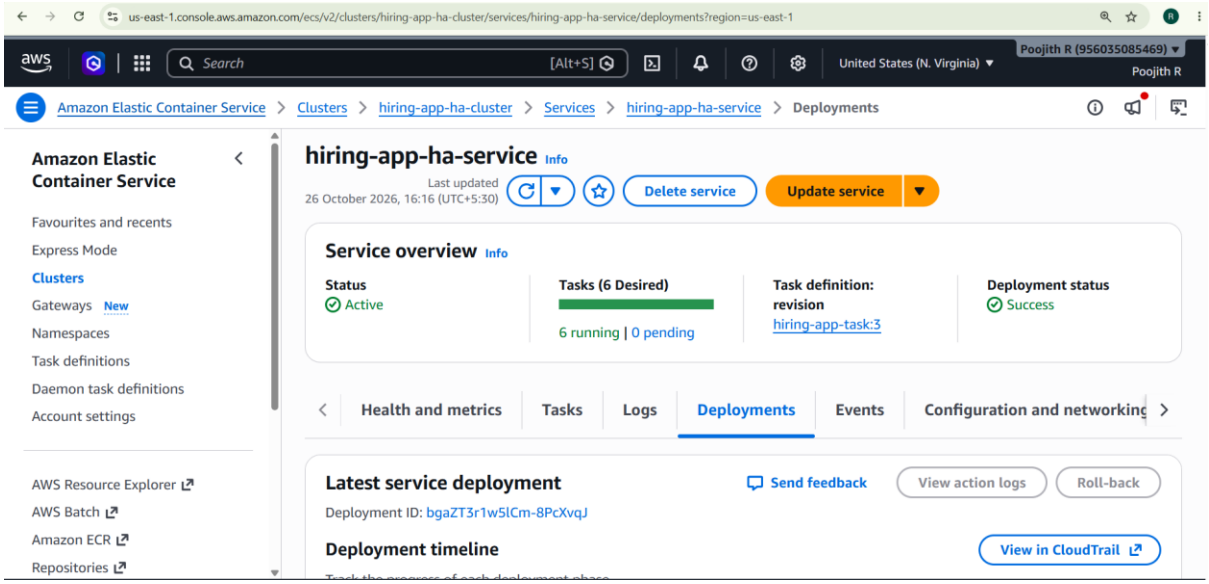

us-east-1.console.aws.amazon.com/ecs/v2/task-definitions/hiring-app-task/3/containers
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service > Task definitions > hiring-app-task > Revision 3 > Containers
Amazon Elastic Container Service
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
Task definition successfully created
hiring-app-task:3 has been successfully created. You can use this task definition to deploy a service or run a task.
View task definition
hiring-app-task:3
Last updated 26 October 2026, 16:03 (UTC+5:30)
Deploy
Actions
Create new revision
Overview Info
ARN
arn:aws:ecs:us-east-1:956035085469:task-definition/hiring-app-task:3
Status
ACTIVE
Time created
26 October 2026, 16:03 (UTC+5:30)
App environment
EC2
Task role
ecsTaskExecutionRole
Task execution role
ecsTaskExecutionRole
Operating system/Architecture
Linux/X86_64
Network mode
bridge
Fault injection
Turned off
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
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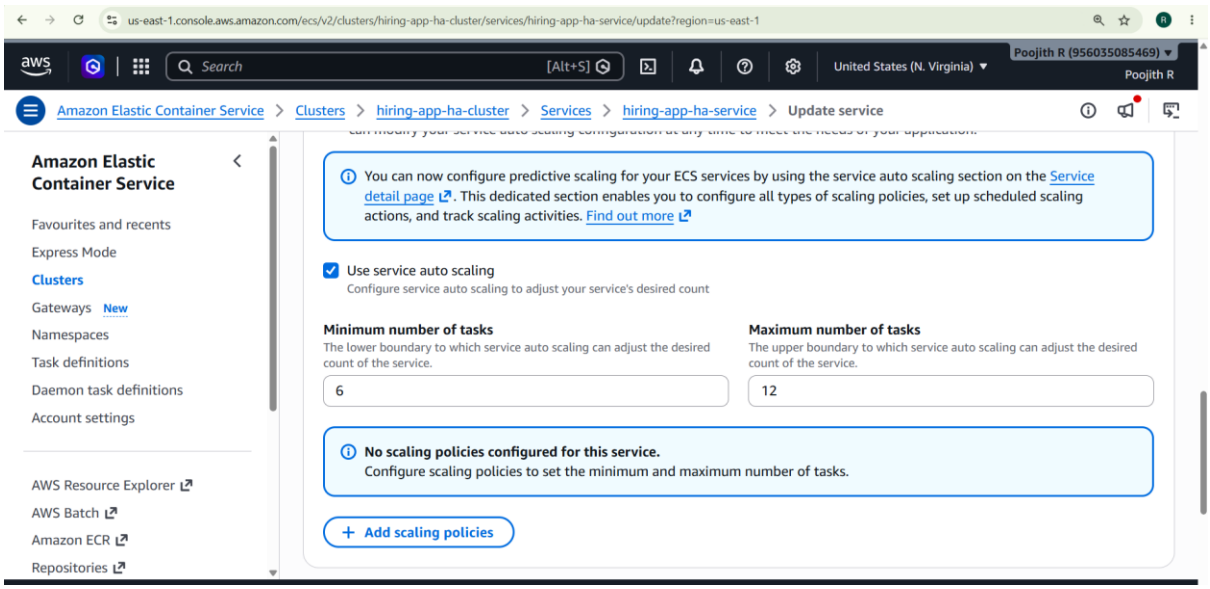

us-east-1.console.aws.amazon.com/ecs/v2/clusters/hiring-app-ha-cluster/services?region=us-east-1
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service
Clusters
hiring-app-ha-cluster
Services
Amazon Elastic Container Service
hiring-app-ha-service deployment is in progress. It takes a few minutes.
View in CloudFormation
Notifications 0 0 1 0 1
hiring-app-ha-cluster
ASG
Last updated
26 October 2026, 16:11 (UTC+5:30)
Actions
Create with Express Mode
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
Cluster overview
Cluster ARN
arn:aws:ecs:us-east-1...
Status
Active
CloudWatch monitoring
Default
Action logs
Turned off
Services
0 active | 0 draining
Tasks
0 running | 0 pending
Registered container instances
2
Services
Tasks
Daemons
Infrastructure – updated
Metrics
Scheduled tasks
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
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us-east-1.console.aws.amazon.com/ecs/v2/clusters/hiring-app-ha-cluster/services?region=us-east-1
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service
Clusters
hiring-app-ha-cluster
Services
Amazon Elastic Container Service
hiring-app-ha-service has been deployed successfully.
View service
hiring-app-ha-cluster
ASG
Last updated
26 October 2026, 16:11 (UTC+5:30)
Actions
Create with Express Mode
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
Cluster overview
Cluster ARN
arn:aws:ecs:us-east-1...
Status
Active
CloudWatch monitoring
Default
Action logs
Turned off
Services
0 active | 0 draining
Tasks
0 running | 0 pending
Registered container instances
2
Services
Tasks
Daemons
Infrastructure – updated
Metrics
Scheduled tasks


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#TargetGroups:
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
EC2
Target groups
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Target groups (1/1) Info | What's new?
Actions
Create target group
Filter target groups
Name
ARN
Port
Protocol
Target type
hiring-app-tg
arn:aws:elasticloadbalancin...
8080
HTTP
Instance
Target group: hiring-app-tg
Filter targets
Instance ID
Name
Port
Zone
Health status
i-03055116d7a5a40cd
ECS Instance - ...
32768
us-east-1b (us...
Healthy


us-east-1.console.aws.amazon.com/ecs/v2/clusters/hiring-app-ha-cluster/services/hiring-app-ha-service/deployments?region=us-east-1
Search [Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service > Clusters > hiring-app-ha-cluster > Services > hiring-app-ha-service > Deployments
Amazon Elastic Container Service
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
Service updated: hiring-app-ha-cluster:hiring-app-ha-service
hiring-app-ha-service Info
Last updated
26 October 2026, 16:15 (UTC+5:30)
Delete service
Update service
Service overview Info
Status
Active
Tasks (6 Desired)
1 running | 0 pending
Task definition: revision
hiring-app-task:3
Deployment status
Success
Health and metrics
Tasks
Logs
Deployments
Events
Configuration and networking
Latest service deployment
Send feedback
View action logs
Roll-back
Deployment ID: bgaZT3r1w5lCm-8PcXvqJ
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
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us-east-1.console.aws.amazon.com/ecs/v2/clusters/hiring-app-ha-cluster/services/hiring-app-ha-service/deployments?region=us-east-1
Search [Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service > Clusters > hiring-app-ha-cluster > Services > hiring-app-ha-service > Deployments
Amazon Elastic Container Service
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
hiring-app-ha-service Info
Last updated
26 October 2026, 16:16 (UTC+5:30)
Delete service
Update service
Service overview Info
Status
Active
Tasks (6 Desired)
6 running | 0 pending
Task definition: revision
hiring-app-task:3
Deployment status
Success
Health and metrics
Tasks
Logs
Deployments
Events
Configuration and networking
Latest service deployment
Send feedback
View action logs
Roll-back
Deployment ID: bgaZT3r1w5lCm-8PcXvqJ
Deployment timeline
View in CloudTrail


us-east-1.console.aws.amazon.com/ecs/v2/clusters/hiring-app-ha-cluster/services/hiring-app-ha-service/update?region=us-east-1
Search [Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Amazon Elastic Container Service > Clusters > hiring-app-ha-cluster > Services > hiring-app-ha-service > Update service
Amazon Elastic Container Service
Favourites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
You can now configure predictive scaling for your ECS services by using the service auto scaling section on the Service detail page. This dedicated section enables you to configure all types of scaling policies, set up scheduled scaling actions, and track scaling activities. Find out more
Use service auto scaling
Configure service auto scaling to adjust your service's desired count
Minimum number of tasks
The lower boundary to which service auto scaling can adjust the desired count of the service.
6
Maximum number of tasks
The upper boundary to which service auto scaling can adjust the desired count of the service.
12
No scaling policies configured for this service.
Configure scaling policies to set the minimum and maximum number of tasks.
Add scaling policies



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